

Previous Year Questions

Some Basic Concepts of Chemistry

Short Answer (2 Marks)

Q1: What is Biology, and what does the living world comprise? (2 Marks)

- Biology is defined as the science of life forms and living processes.
- The living world is composed of an extraordinary variety of living organisms.

Q2: What is biodiversity, and what is its known range? (2 Marks)

- Biodiversity refers to the number and types of organisms present on Earth.
- The number of known and described species ranges between 1.7 to 1.8 million.

Q3: Define nomenclature and identification in the context of living organisms. (2 Marks)

- Nomenclature is the process of standardizing the naming of living organisms so that a particular organism is known by the same name globally.
- Identification is the process of correctly describing an organism to know which organism a name is attached to, making nomenclature possible.

Q4: What is Binomial Nomenclature, and who proposed it? (2 Marks)

- Binomial nomenclature is a system of providing a scientific name with two components: the Generic name and the specific epithet.
- This naming system was given by Carolus Linnaeus.

Q5: What are the two components of a biological name, using the example of mango? (2 Marks)

- In the scientific name of mango, 'Mangifera indica', 'Mangifera' represents the genus.
- While 'indica' is the specific epithet, indicating a particular species.

Q6: What are the first two universal rules of nomenclature regarding biological names? (2 Marks)

- Biological names are generally in Latin and are written in italics.

- They are Latinised or derived from Latin, regardless of their original language.

Q7: What is the rule for writing the first word and specific epithet in a biological name? (2 Marks)

- The first word, representing the genus, starts with a capital letter.
- The specific epithet, the second component, starts with a small letter.

Q8: What is classification, and why is it necessary? (2 Marks)

- Classification is the process by which anything is grouped into convenient categories based on some easily observable characters.
- It is necessary because it is almost impossible to study all living organisms individually, making systematic grouping essential.

Q9: What is a 'taxon'? Give an example. (2 Marks)

- The scientific term for the convenient categories we use to study organisms is taxa (singular: taxon).
- Examples include 'Plants', 'Animals', 'Mammals', and 'Dogs', all of which represent different taxa.

Q10: What is taxonomy, and what processes are basic to it? (2 Marks)

- Taxonomy is the process of classifying all living organisms into different taxa based on characteristics.
- The basic processes of taxonomy are characterisation, identification, classification, and nomenclature.

Medium Answer (3 Marks)

Q11: Discuss the relationship between early human societies and biological knowledge, and how systematic description of life forms led to progress. (3 Marks)

- Early humans could easily perceive the difference between inanimate matter and living organisms.
- They deified some of the inanimate matter (wind, sea, fire etc.) and some among the animals and plants.

- Societies which indulged in anthropocentric view of biology could register limited progress in biological knowledge.
- Systematic and monumental description of life forms brought in, out of necessity, detailed systems of identification, nomenclature and classification.

Q12: Explain the significance of the recognition of similarities among living organisms in the context of biodiversity. (3 Marks)

- The biggest spin off of such studies was the recognition of the sharing of similarities among living organisms both horizontally and vertically.
- This revelation that all present day living organisms are related to each other and also to all organisms that ever lived on this earth, humbled man.
- This understanding led to cultural movements for conservation of biodiversity.

Q13: Describe Ernst Mayr's contributions to evolutionary biology and his notable achievements. (3 Marks)

- Ernst Mayr (1904-2004), the Harvard University evolutionary biologist, has been called 'The Darwin of the 20th century' and was one of the 100 greatest scientists of all time.
- Throughout his nearly 80-year career, his research spanned ornithology, taxonomy, zoogeography, evolution, systematics, and the history and philosophy of biology.
- He almost single-handedly made the origin of species diversity the central question of evolutionary biology that it is today.
- He also pioneered the currently accepted definition of a biological species.
- Mayr was awarded the three prizes widely regarded as the triple crown of biology: the Balzan Prize in 1983, the International Prize for Biology in 1994, and the Crafoord Prize in 1999.

Q14: Explain the two implicit questions within the reflection on 'what indeed is life?' according to scientists. (3 Marks)

- The question 'what indeed is life?' has two implicit questions within it.
- The first is a technical one, seeking an answer to what living is as opposed to the non-living.
- The second is a philosophical one, seeking an answer to what the purpose of life is.

- As scientists, the text states that they shall not attempt answering the second (philosophical) question, but will try to reflect on 'what is living'.

Q15: Why is a standardized naming system for living organisms necessary, and what organizations are involved in this process for plants and animals? (3 Marks)

- Local names for plants and animals vary from place to place, even within a country, which would create confusion if there were no standardized ways to refer to them.
- Hence, there is a need to standardise the naming of living organisms such that a particular organism is known by the same name all over the world.
- For plants, scientific names are based on agreed principles and criteria provided in the International Code for Botanical Nomenclature (ICBN).
- For animals, animal taxonomists have evolved the International Code of Zoological Nomenclature (ICZN).

Q16: Elaborate on the universal rules of nomenclature regarding the formatting and authorship of scientific names. (3 Marks)

- Biological names are generally in Latin and written in italics, or they are Latinised or derived from Latin irrespective of their origin.
- The first word in a biological name represents the genus while the second component denotes the specific epithet.
- Both words in a biological name, when handwritten, are separately underlined, or printed in italics, to indicate their Latin origin.
- The first word denoting the genus starts with a capital letter while the specific epithet starts with a small letter, as illustrated by **Mangifera indica**.
- The name of the author appears after the specific epithet, i.e., at the end of the biological name and is written in an abbreviated form, for example, **Mangifera indica** Linn., indicating that this species was first described by Linnaeus.

Q17: Explain the concept of 'taxa' with suitable examples, demonstrating how they can indicate categories at different levels. (3 Marks)

- The scientific term for convenient categories we use to study organisms is taxa.
- Taxa can indicate categories at very different levels.

- For example, 'Plants' form a taxa, and 'Wheat' is also a taxa within 'Plants'.
- Similarly, 'animals', 'mammals', and 'dogs' are all taxa, but 'animals', 'mammals', and 'dogs' represent taxa at different levels, where a dog is a mammal and mammals are animals.

Q18: How do taxonomical studies contribute to various fields beyond basic classification? (3 Marks)

- Taxonomic studies of various species of plants and animals are useful in agriculture.
- They are also useful in forestry and industry.
- In general, they are beneficial for knowing our bio-resources and their diversity.

Q19: What is a species according to taxonomic studies? How are closely related species distinguished? (3 Marks)

- Taxonomic studies consider a group of individual organisms with fundamental similarities as a species.
- One should be able to distinguish one species from other closely related species based on distinct morphological differences.
- For example, *Mangifera indica*, *Solanum tuberosum* (potato), and *Panthera leo* (lion) are distinct species, with *indica*, *tuberosum*, and *leo* representing the specific epithets.
- *Panthera tigris* is another specific epithet for the genus *Panthera*, and *Solanum* includes species like *nigrum* and *melongena*.
- Human beings belong to the species *sapiens*, grouped in the genus *Homo*, making their scientific name *Homo sapiens*.

Q20: Describe the taxonomic category of 'Genus' with an example. (3 Marks)

- Genus comprises a group of related species which has more characters in common in comparison to species of other genera.
- Genera are aggregates of closely related species.
- For example, potato and brinjal are two different species but both belong to the genus *Solanum*.

- Lion (*Panthera leo*), leopard (*P. pardus*) and tiger (*P. tigris*) with several common features, are all species of the genus *Panthera*.
- This genus *Panthera* differs from another genus *Felis*, which includes cats.

Long Answer (5 Marks)

Q21: Explain the concept of taxonomic hierarchy. Illustrate with an example and discuss how common characteristics change as one moves up or down the hierarchy. (5 Marks)

- Classification is not a single step process but involves a hierarchy of steps in which each step represents a rank or category.
- Since the category is a part of overall taxonomic arrangement, it is called the taxonomic category and all categories together constitute the taxonomic hierarchy.
- Each category, referred to as a unit of classification, in fact, represents a rank and is commonly termed as taxon (pl.: taxa).
- Taxonomical studies of all known organisms have led to the development of common categories such as kingdom, phylum or division (for plants), class, order, family, genus and species. Species is the lowest category for all organisms.
- For example, considering the classification of humans (Man):
 - * **Species**: *sapiens*
 - * **Genus**: *Homo*
 - * **Family**: Hominidae
 - * **Order**: Primata
 - * **Class**: Mammalia
 - * **Phylum**: Chordata
 - * **Kingdom**: Animalia
- As we go higher from species to kingdom, the number of common characteristics goes on decreasing. Lower the taxa, more are the characteristics that the members within the taxon share. Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level, making the problem of classification more complex.

Q22: Describe the taxonomic categories of 'Family' and 'Order'. Provide examples for both plants and animals to illustrate their characteristics and relationships. (5 Marks)

- **Family:** The next category, Family, has a group of related genera with still less number of similarities as compared to genus and species. Families are characterised on the basis of both vegetative and reproductive features of plant species.
- **Plant Example:** Among plants, for example, three different genera *Solanum*, *Petunia* and *Datura* are placed in the family Solanaceae.
- **Animal Example:** Among animals, for example, genus *Panthera*, comprising lion, tiger, leopard is put along with genus, *Felis* (cats) in the family Felidae. If you observe the features of a cat and a dog, you will find some similarities and some differences as well, and they are separated into two different families - Felidae and Canidae, respectively.
- **Order:** Order being a higher category, is the assemblage of families which exhibit a few similar characters. The similar characters are less in number as compared to different genera included in a family. Generally, order and other higher taxonomic categories are identified based on the aggregates of characters.
- **Plant Example:** Plant families like Convolvulaceae, Solanaceae are included in the order Polymoniales mainly based on the floral characters.
- **Animal Example:** The animal order, Carnivora, includes families like Felidae and Canidae.

Q23: Describe the taxonomic categories of 'Class' and 'Phylum'/'Division'. Explain how organisms are grouped at these higher levels. (5 Marks)

- **Class:** This category includes related orders.
- **Example:** For example, order Primata comprising monkey, gorilla and gibbon is placed in class Mammalia along with order Carnivora that includes animals like tiger, cat and dog. Class Mammalia has other orders also.
- **Phylum (for animals) or Division (for plants):** Classes comprising animals like fishes, amphibians, reptiles, birds along with mammals constitute the next higher category called Phylum.
- **Animal Example:** All these, based on the common features like presence of notochord and dorsal hollow neural system, are included in phylum Chordata.

- *** **Plant Example**:** In case of plants, classes with a few similar characters are assigned to a higher category called Division.

Q24: Discuss the highest taxonomic category, 'Kingdom', and how it differentiates the major groups of life. Also, explain the challenges in classification as one moves to higher categories. (5 Marks)

- ****Kingdom:**** All animals belonging to various phyla are assigned to the highest category called Kingdom Animalia in the classification system of animals. The Kingdom Plantae, on the other hand, is distinct, and comprises all plants from various divisions. Henceforth, these two groups are referred to as animal and plant kingdoms.
- ****Challenges in Classification at Higher Categories:**** The taxonomic categories from species to kingdom are shown in ascending order. As we go higher from species to kingdom, the number of common characteristics goes on decreasing. Lower the taxa, more are the characteristics that the members within the taxon share. Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level. Hence, the problem of classification becomes more complex.

Q25: Elaborate on the early history of human interest in classifying organisms, highlighting their motivations and the evolution of systematic study. (5 Marks)

- Human beings have always been interested in knowing more and more about the various kinds of organisms, particularly with reference to their own use.
- In early days, human beings needed to find sources for their basic needs of food, clothing and shelter. Hence, the earliest classifications were based on the 'uses' of various organisms.
- Societies which indulged in anthropocentric view of biology could register limited progress in biological knowledge.
- Human beings were, since long, not only interested in knowing more about different kinds of organisms and their diversities, but also the relationships among them.
- This branch of study was referred to as systematics. The word systematics is derived from the Latin word 'systema' which means systematic arrangement of organisms. Linnaeus used **Systema Naturae** as the title of his publication. The scope of systematics was later enlarged to include identification, nomenclature and classification. Systematics also takes into account evolutionary relationships between organisms.

Q26: Explain how taxonomical categories and hierarchy can be illustrated using the example of 'Insects'. (5 Marks)

- Taxonomic categories and hierarchy can be illustrated by an example.
- Insects represent a group of organisms sharing common features like three pairs of jointed legs.
- It means insects are recognisable concrete objects which can be classified, and thus were given a rank or category.
- This group represents a category, and category further denotes rank. Each rank or taxon, in fact, represents a unit of classification.
- These taxonomic groups/categories are distinct biological entities and not merely morphological aggregates.
- The basic requirement to place an organism in various categories is the knowledge of characters of an individual or group of organisms. This helps in identifying similarities and dissimilarities among the individuals of the same kind of organisms as well as of other kinds of organisms.

Q27: Provide a detailed explanation of the taxonomic classification for 'Man' and 'Housefly', including all categories from Kingdom to Species. (5 Marks)

- The taxonomic classification of Man and Housefly is as follows:
- **Man:**
- **Common Name:** Man
- **Biological Name:** *Homo sapiens*
- **Genus:** *Homo* (Species *sapiens* is grouped in genus *Homo*).
- **Family:** Hominidae
- **Order:** Primata (Comprising monkey, gorilla and gibbon, placed in class Mammalia).
- **Class:** Mammalia (Includes order Primata and Carnivora).
- **Phylum:** Chordata (Includes classes like fishes, amphibians, reptiles, birds, and mammals, based on common features like presence of notochord and dorsal hollow neural

system).

- * **Kingdom:** Animalia (Highest category for animals).
- **Housefly:**
- * **Common Name:** Housefly
- * **Biological Name:** *Musca domestica*
- * **Genus:** *Musca*
- * **Family:** Muscidae
- * **Order:** Diptera
- * **Class:** Insecta
- * **Phylum:** Arthropoda
- * **Kingdom:** Animalia

Q28: Provide a detailed explanation of the taxonomic classification for 'Mango' and 'Wheat', including all categories from Kingdom to Species. (5 Marks)

- The taxonomic classification of Mango and Wheat is as follows:
- **Mango:**
- * **Common Name:** Mango
- * **Biological Name:** *Mangifera indica* (*indica* is the specific epithet, *Mangifera* is the genus).
- * **Genus:** *Mangifera*
- * **Family:** Anacardiaceae
- * **Order:** Sapindales
- * **Class:** Dicotyledonae
- * **Division:** Angiospermae (For plants, classes with a few similar characters are assigned to a higher category called Division).
- * **Kingdom:** Plantae (Distinct, comprising all plants from various divisions).
- **Wheat:**

- * **Common Name:** Wheat
- * **Biological Name:** *Triticum aestivum*
- * **Genus:** *Triticum*
- * **Family:** Poaceae
- * **Order:** Poales
- * **Class:** Monocotyledonae
- * **Division:** Angiospermae
- * **Kingdom:** Plantae

Q29: Discuss the process of identification and its crucial role in biological studies, linking it to nomenclature and classification. (5 Marks)

- Identification is a crucial process where an organism is described correctly to know to what organism the name is attached. It is a necessary prerequisite for nomenclature and classification.
- The need for identification arises because local names vary widely, causing confusion in communication about organisms globally.
- To facilitate study, scientists have established procedures to assign scientific names, which is acceptable to biologists worldwide.
- The basic requirement for placing an organism in various categories is the knowledge of characters of an individual or group of organisms. This knowledge helps in identifying similarities and dissimilarities among individuals of the same kind and other kinds of organisms.
- Once an organism is identified, nomenclature ensures that each organism has only one name globally, and such a name has not been used for any other known organism. This process is based on universally accepted principles for providing scientific names.
- Finally, classification is the process by which anything is grouped into convenient categories based on some easily observable characters, a process made possible by accurate identification. Hence, characterisation, identification, classification and nomenclature are the processes that are basic to taxonomy.

Q30: What is systematics, and how has its scope evolved over time? How does it differ from taxonomy? (5 Marks)

- Systematics is a branch of study that was referred to as the study of knowing more about different kinds of organisms and their diversities, and also the relationships among them. The word systematics is derived from the Latin word 'systema' which means systematic arrangement of organisms. Linnaeus used *Systema Naturae* as the title of his publication.
- ****Evolution of Scope:**** The scope of systematics was later enlarged to include identification, nomenclature and classification. Crucially, systematics now also takes into account evolutionary relationships between organisms. This evolution shows a shift from mere arrangement to understanding the historical and evolutionary connections between life forms.
- ****Difference from Taxonomy:**** While closely related and often used in conjunction, systematics and taxonomy have distinct focuses:
- * ****Taxonomy**** is the process of classification based on characteristics, encompassing characterisation, identification, classification, and nomenclature. It's about devising means to make the study of living organisms possible through grouping them into convenient categories.
- * ****Systematics**** has a broader scope. While it includes the principles and procedures of taxonomy (identification, nomenclature, classification), its unique aspect is the consideration of evolutionary relationships among organisms. Thus, systematics aims to understand the diversity of life in an evolutionary context, providing a framework for the classification developed by taxonomy.